

**National University of Computer and Emerging Sciences,
Lahore Campus**

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Software Engineering)	Semester:	Spring 2026
	Duration:	20 minutes	Total Marks:	15
	Paper Date:	05-May-2026	Section	6A , 6B
	Exam Type:	Quiz 6 - Chapter 6	Page(s):	2

Student Name

Roll No.

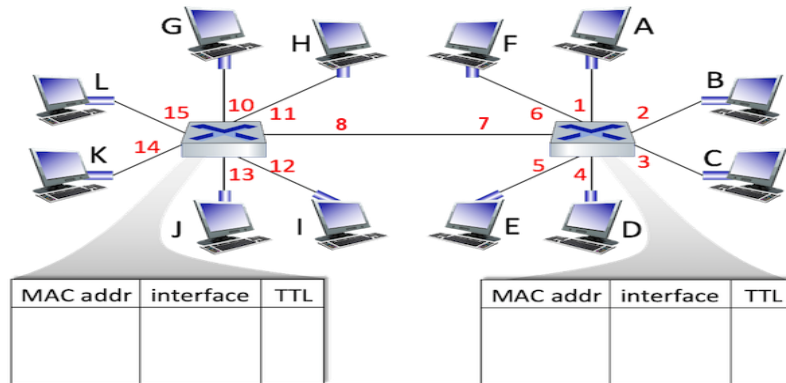
Section:

Q1. Encircle the correct option:

[5 Marks] [CLO 3]

- The DHCP Request message is encapsulated in which Transport Layer protocol?
 - TCP
 - UDP
 - ARP
 - It is not encapsulated in a Transport Layer; it sits directly on IP.
- After the DHCP handshake, the laptop sends a DNS query for www.google.com. The DNS Server is on a remote network. The frame containing this DNS query has which Destination MAC address?
 - The MAC address of the Google Web Server.
 - The MAC address of the DNS Server.
 - The MAC address of the First-Hop Router.
 - FF:FF:FF:FF:FF:FF
- A CRC system uses generator polynomial $G(x) = 1101$. Which of the following error patterns is guaranteed to be detected?
 - 1100
 - 1010
 - 1000
 - 111
- Which of the following generators satisfies both conditions: (1) reliably detects all single-bit errors, AND (2) reliably detects all odd-numbered bit errors?
 - 1000 (x^3)
 - 111 ($x^2 + x + 1$)
 - 1001 ($x^3 + 1$)
 - None
- A layer 2 switch can use:
 - flooding
 - selective forwarding
 - both a & b
 - none

Q2. Consider the LAN below consisting of 10 computers connected by two self-learning Ethernet switches. At $t=0$ the switch table entries for both switches are empty. At $t=1$ and 2 , a source node sends to a destination node as shown below, and the destination replies immediately (well before the next time step). **[10 Marks] [CLO 3]**



Assume that the following transmissions occur:

$t=1$: $H \rightarrow A$

$t=2$: $K \rightarrow I$

Please answer the following questions:

1. At $t=1$, what is the source entry for switch 1?
Answer: _____
2. At $t=1$, what is the destination entry for switch 1?
Answer: _____
3. At $t=1$, what is the source entry for switch 2?
Answer: _____
4. At $t=1$, what is the destination entry for switch 2?
Answer: _____
5. At $t=2$, what is the source entry for switch
Answer: _____
6. At $t=2$, what is the destination entry for switch 1?
Answer: _____
7. At $t=2$, what is the source entry for switch 2?
Answer: _____
8. At $t=2$, what is the destination entry for switch 2?
Answer: _____
9. What is the physical address of interface 8 of the switch at the left side in the diagram?
Answer: _____
10. What is the physical address of interface 7 of the switch at the right side in the diagram?
Answer: _____